

Part B: GenAI-Assisted Critique

Prompt 1: Protocol Design & Confounding Variables Critique

I have designed a sensor protocol for an anti-theft desk security system to differentiate between normal desk activities and actual theft while reducing false alarms. I am using a PIR motion sensor (body heat) and an LSM6DS3 accelerometer (3-axis motion/shaking) on an Arduino Nano 33 IoT board, sampling at 1 Hz for 24 hours. My hypotheses are:

1. *H1*: Combining PIR and accelerometer data reduces false alarm rates to <5% while maintaining >95% theft detection compared to single-sensor setups.
2. *H2*: Transient desk bumps last <1 second, whereas actual theft movement persists across consecutive readings (>2–3 seconds), which can be filtered out using a moving average / duration threshold.

What are potential weaknesses in this experimental design? What confounding environmental or physical variables (e.g., HVAC airflow, table movement, ambient vibration) might I have missed? Suggest 2–3 methodological improvements.

Prompt 2: Ethical & Privacy Gap Identification

I am deploying an anti-theft security device using PIR motion and accelerometer sensors in a home/office environment. My privacy protocol includes: logging non-biometric raw numbers (Timestamp, PIR_State, Accel_X/Y/Z, Accel_Delta) at 1 Hz, retaining raw CSVs locally for 7 days, aggregating data into daily summaries for up to 90 days, and displaying a physical privacy notice.

What ethical, power dynamic, or privacy gaps might I have overlooked regarding room occupants or family members? Who else could be affected by this data

collection? Suggest one additional safeguard to align with Privacy-by-Design principles (ISO 27001).

Prompt 3: Literature & Methodological Soundness

I identified a gap in current anti-theft research: existing smartphone/CCTV setups often produce high false-positive rates due to environmental noise or lack multi-sensor verification without continuous cloud dependence. My project tests Multi-Sensor Data Fusion (MSDF) combining PIR spatial detection and IMU acceleration magnitude at 1 Hz.

What statistical assumptions (e.g., temporal autocorrelation, idle noise normality) should I verify for 1 Hz time-series sensor data? Are there established data fusion frameworks or key papers in IoT physical security I should cite to justify my design?

Prompt one result Summary: Protocol Design & Confounding Variables Critique

Key Design Weaknesses

1. **Sampling Rate Too Slow (1 Hz):** Fast thefts (200–500 ms) will be missed or look identical to a table bump, breaking H_2 .
2. **PIR Sensor Blindspots:** HVAC/sunlight cause false positives; stationary thieves cause false negatives, threatening H_1 's >95% accuracy claim.
3. **Uncontrolled Confounders:** Fails to isolate signal from ambient vibration (footsteps/traffic), table resonance, thermal drift, and WiFi EMI.
4. **Vague "Theft" Definition:** Lacks precise physical metrics (speed, weight, distance) for what actually constitutes theft.

Key Methodological Fixes

- **Boost Sampling to 10–20+ Hz:** Enables frequency filtering (FFT) to easily separate quick bumps from sustained lifting.
- **Deploy a Sensor Array:** Uses multiple accelerometers across the desk to distinguish localized theft from whole-table vibrations.
- **48-Hour Stratified Baseline:** Logs environmental noise and HVAC cycles to create adaptive trigger thresholds.

Revised Hypotheses (In Short)

Refined H_1 : Multi-sensor fusion at 10+ Hz with spatial filtering drops false alarms <5% across all environmental cycles.

Refined H_2 : Theft produces localized, sustained (>0.5s) 1–3 Hz movement; bumps produce brief (<300 ms) broadband spikes.

Prompt Two result Summary: Ethical & Privacy Gap Identification

Key Privacy & Ethical Gaps

1. **Consent & Power Dynamics:** A static privacy notice lacks ongoing consent, withdrawal mechanisms, or protection for non-consenting occupants (e.g., family, guests, co-workers who feel pressured).
2. **Scope Creep & Inference Risk:** 1 Hz PIR and accelerometer logs reveal daily routines, presence, and absence patterns—risking secondary misuse like behavior or productivity tracking.
3. **Arbitrary Data Retention:** The 7-day raw / 90-day aggregate timeline lacks a documented data-minimization rationale; a shorter timeframe (e.g., 48 hours raw) would significantly reduce risk.
4. **Overlooked Stakeholders:** Guests, renters, family members, and third parties (e.g., landlords, insurers) could be impacted without direct consent.

Key Privacy-by-Design Safeguard

Role-Based Access Control (RBAC) & Audit Logging:

- **Device Owner:** Access to real-time alerts and short-term raw logs only.
- **Facility/Room Manager:** Access restricted strictly to high-level daily aggregates.
- **Audit Trail:** Log every access request (timestamp, user, purpose) to ensure accountability, purpose limitation, and transparency per **ISO 27001** standards.

Prompt Three result Summary: Literature & Methodological Soundness

Key Statistical Assumptions to Verify

1. **Temporal Autocorrelation:** 1 Hz samples are not independent. Run **ACF / Ljung-Box** tests to avoid inflating statistical significance.
2. **Stationarity & Noise:** Check for baseline drift (temperature/orientation) using **ADF tests**. Verify if idle noise is Gaussian (**Shapiro-Wilk**) to justify standard $\mu + 3\sigma$ thresholds.
3. **Sensor Lag & Independence:** Use **cross-correlation** to account for time lags between PIR heat detection and IMU shaking, and test for conditional independence before applying Naïve Bayes.

Critical Evaluation Metrics

- **Prioritize False Alarms:** Overall accuracy is misleading for security. Report **False Positive Rate (FPR)**, **False Alarms per Hour**, and **Detection Latency**.
- **Standard Classifiers:** Include **Precision**, **Recall**, **F1-Score**, and **ROC-AUC** to prove system reliability.

Fusion Frameworks & Core Literature

Theoretical Frameworks:

- **Dempster-Shafer Evidence Theory:** Best for handling sensor conflicts and uncertainty (e.g., PIR false triggers vs. accidental bumps).
- **Bayesian Data Fusion:** Provides lightweight, edge-friendly probabilistic decision-making.
- **JDL Model:** Frame your system as Feature-level or Decision-level fusion.

Key Citations to Include:

- **Hamda et al. (2023):** Justifies Dempster-Shafer theory for uncertainty in IoT multisensor fusion.
- **Yuan et al. (2022):** Supports PIR/multimodal fusion as a privacy-preserving alternative to continuous video.
- **Hall & Llinas:** Classic foundational citation for Multi-Sensor Data Fusion (MSDF).

Concise Refined Hypothesis

H1: A multi-sensor data fusion framework combining PIR motion detection and IMU acceleration magnitude at 1 Hz significantly reduces false-positive theft alerts compared with single-sensor approaches by requiring corroborating evidence from independent sensing modalities.

GenAI feedback summary

Prompt / Topic	GenAI Response (key points)	Tool used	Date accessed
Protocol design critique	1. Sampling Rate Too Slow (1 Hz): Fast thefts (200–500 ms) will be missed or look identical to a table bump, breaking H_2. 2. PIR Sensor Blindspots: HVAC/sunlight cause false positives; stationary thieves cause false negatives, threatening H_1's >95% accuracy claim. 3. Uncontrolled Confounders: Fails to isolate signal from ambient vibration (footsteps/traffic), table resonance, thermal drift, and WiFi EMI. 4. Vague "Theft" Definition: Lacks precise physical metrics (speed, weight, distance) for what actually constitutes theft.	Deakin GEM	August 2026
Ethical gaps	1. Consent & Power Dynamics: A static privacy notice lacks ongoing consent, withdrawal mechanisms, or protection for non-consenting occupants (e.g., family, guests, co-workers who feel pressured). 2. Scope Creep & Inference Risk: 1 Hz PIR and accelerometer logs reveal daily routines, presence, and absence patterns—risking secondary misuse like behavior or productivity tracking. 3. Arbitrary Data Retention: The 7-day raw / 90-day aggregate timeline lacks a documented data-minimization rationale; a shorter timeframe (e.g., 48 hours raw) would significantly reduce risk. 4. Overlooked Stakeholders: Guests, renters, family members, and third parties (e.g., landlords, insurers) could be impacted without	Deakin GEM	August 2026

	direct consent.		
Literature validation	1. Temporal Autocorrelation: 1 Hz samples are not independent. Run ACF / Ljung-Box tests to avoid inflating statistical significance. 2. Stationarity & Noise: Check for baseline drift (temperature/orientation) using ADF tests. Verify if idle noise is Gaussian (Shapiro-Wilk) to justify standard $\mu + 3\sigma$ thresholds. 3. Sensor Lag & Independence: Use cross-correlation to account for time lags between PIR heat detection and IMU shaking, and test for conditional independence before applying Naïve Bayes.	Copilot	August 2026

I have designed a sensor protocol for an anti-theft desk security system to differentiate between normal desk activities and actual theft while reducing false alarms. I am using a PIR motion sensor (body heat) and an LSM6DS3 accelerometer (3-axis motion/shaking) on an Arduino Nano 33 IoT board, sampling at 1 Hz for 24 hours. My hypotheses are:

- H1: Combining PIR and accelerometer data reduces false alarm rates to <5% while maintaining >95% theft detection compared to single-sensor setups.
- H2: Transient desk bumps last <1 second, whereas actual theft movement persists across consecutive readings (>2-3 seconds), which can be filtered out using a moving average / duration threshold.

What are potential weaknesses in this experimental design? What confounding environmental or physical variables (e.g., HVAC airflow, table movement, ambient vibration) might I have missed? Suggest 2-3 methodological improvements.

Q1 Claude Haiku 4.5: Fast responses with minimal wait time

搜尋到 3 個來源

I'll evaluate your sensor protocol design. This is a solid starting point, but there are real weaknesses to address.

Weaknesses in Current Design

傳送訊息

+ 搜尋

Terms of Use: Deakin GEM is provided to support a range of tasks and activities. By accessing Deakin GEM, you acknowledge and accept that your use is subject to the [Student Rights and Responsibilities \(for students\)](#), the [Code of Conduct \(for staff\)](#), and the [Information and Communications Technology Acceptable Use Policy \(for all users\)](#), along with any related procedures.

Use of responses: Deakin GEM uses artificial intelligence to generate responses. While it aims to provide helpful and relevant information, responses may be incomplete, outdated, or inaccurate and should not be relied on as a sole source of information. Please review and verify information where appropriate.

Weaknesses in Current Design

1. Sampling rate too low for motion discrimination

The problem:

1 Hz (1 sample per second) is insufficient to distinguish a genuine theft event from a transient bump.

- A quick grab or lift might occupy 200–500 ms
- At 1 Hz, you may capture only 1–2 samples during the entire event
- You cannot reliably distinguish duration or acceleration *signature* at this resolution

Impact on H2:

Your threshold of >2–3 seconds requires consecutive readings, but with only 1 sample per second, you have very coarse temporal resolution. A thief grabbing something in 0.5 seconds might appear identical to a bump.

2. PIR sensor limitations underexplored

- **False positives:** HVAC drafts, sunlight changes, and radiator heat can trigger PIR
- **False negatives:** A person standing still (common during theft preparation) won't trigger a body-heat sensor
- **Assumption mismatch:** You're assuming theft *always involves motion*; it often doesn't (person waits, then grabs)

This directly threatens H1's claim of >95% detection.

3. Confounding variables not controlled

Your design doesn't systematically isolate signal from noise:

FACTOR	EFFECT	YOUR CONTROL?	 
HVAC airflow	Pushes lightweight objects; triggers PIR false positives	✗ None mentioned	
Ambient vibration (footsteps, traffic, doors)	Accelerometer noise floor	✗ Not baseline-tested	
Table resonance	A bump on one desk corner radiates to another; accelerometer picks it up globally	✗ Single sensor placement	
Temperature drift	PIR baseline shifts over 24h; sensitivity drifts	✗ Not corrected	
Electromagnetic interference	LSM6DS3 and Arduino on same board; EMI from WiFi (IoT)	✗ No shielding/filtering mentioned	

4. Weak operational definition of "theft"

What counts as theft in your test?

- How fast must an object move?
- What mass?
- Does the object leave the desk, or just get relocated?

Without this, H1 and H2 become unfalsifiable.

Methodological Improvements

Improvement 1: Increase sampling rate + add spectral filtering

Change:

- Sample accelerometer at **10–20 Hz** (minimum; 50 Hz preferred)
- Keep PIR at 1 Hz if power-constrained, but timestamp it precisely

Why:



Methodological Improvements

Improvement 1: Increase sampling rate + add spectral filtering

Change:

- Sample accelerometer at **10–20 Hz** (minimum; 50 Hz preferred)
- Keep PIR at 1 Hz if power-constrained, but timestamp it precisely

Why:

- You can now resolve acceleration shape: a bump is a sharp spike; theft is typically a sustained ramp or multiple peaks
- Compute **frequency domain features** (FFT) on 2–5 second windows:
 - Theft movements often cluster in 1–3 Hz band (deliberate lifting)
 - Environmental vibration typically >5 Hz (traffic, HVAC harmonics)
- Apply a high-pass filter (>0.5 Hz) to remove drift; notch filter at 50/60 Hz for power-line coupling

Outcome: Separates genuine motion signatures from environmental noise.

Improvement 2: Multi-location accelerometer array + spatial filtering

Change:

- Deploy **3 accelerometers** at different desk locations (corner, center, opposite corner)
- Cross-correlate signals across sensors

Why:

- **Local vs. global motion:** A thief lifting an object from corner A shows high local acceleration there but coordinated motion across sensors
- A bump or vibration affecting the whole desk shows *synchronized* acceleration everywhere
- Spatial correlation distinguishes **localized force** (theft) from **distributed disturbance** (HVAC, footsteps)

Outcome: Reduces false positives from ambient vibration; improves specificity of H1.

Improvement 3: Controlled baseline + environmental stratification

Change:

- Before testing, collect **48-hour baseline** data under controlled conditions:
 - Normal office activity (no theft attempts)
 - HVAC on/off cycles measured separately
 - Calibrated thermal drift (PIR sensitivity over time)
- Build a **per-condition model**:
 - Morning (cool, minimal HVAC)
 - Afternoon (heated, active HVAC)
 - Night (stable)

Why:

- Confounds (temperature, ambient vibration) are not *random*—they're systematic and predictable
- By stratifying, you can correct for time-of-day effects and seasonal HVAC patterns
- Your thresholds (duration, amplitude) become **adaptive**, not fixed—much more robust

Outcome: Eliminates systematic bias; makes H1 and H2 testable against real confounds.

Revised Hypotheses (Recommended)

Stronger H1:

"Multi-sensor fusion (PIR + 3-axis accelerometer at 10+ Hz) + spatial filtering reduces false alarm rates below 5% **on controlled test data** compared to single-sensor baselines, when baseline drift and HVAC cycles are accounted for."

Stronger H2:

"Theft events produce sustained acceleration signatures (>0.5 s) in the 1–3 Hz band localized to one desk zone; bumps produce brief (<300 ms) broadband transients. A 2-second moving-average threshold with spatial filtering correctly classifies 90%+ of events."

Quick Checklist

- ☐ Increase accelerometer sampling to ≥ 10 Hz
- ☐ Characterise PIR + accelerometer baseline drift over 48 h
- ☐ Define operational theft protocol (speed, mass, object type)
- ☐ Add spatial accelerometer array or at least rotate sensor placement
- ☐ Test HVAC on/off cycles as separate conditions
- ☐ Compute frequency features (FFT) in addition to amplitude
- ☐ Report false positive and false negative rates separately

This is a realistic system. The tweaks above move you from exploratory testing to a defensible protocol.

Prompt 2 Result: Ethical & Privacy Gap Identification

Deakin GEM | Students

Desk Anti-Theft Sensor Des...

https://gem.deakin.edu.au/c/882319f8-4f28-4547-afb0-bfa83060b19

Claude Haiku 4.5: Fast responses with minimal wait time

I am deploying an anti-theft security device using PIR motion and accelerometer sensors in a home/office environment. My privacy protocol includes: logging non-biometric raw numbers (Timestamp, PIR_State, Accel_X/Y/Z, Accel_Delta) at 1 Hz, retaining raw CSVs locally for 7 days, aggregating data into daily summaries for up to 90 days, and displaying a physical privacy notice.

What ethical, power dynamic, or privacy gaps might I have overlooked regarding room occupants or family members? Who else could be affected by this data collection? Suggest one additional safeguard to align with Privacy-by-Design principles (ISO 27001).

Claude Haiku 4.5: Fast responses with minimal wait time

搜尋到 2 個來源

I need to address this directly: **the provided context is about a LiteLLM/AWS Bedrock bug and is not relevant to your privacy question.** I'm setting it aside and answering from first principles and general privacy knowledge.

Privacy Gaps in Your Protocol

1. Consent and awareness asymmetry

Your physical privacy notice informs people that data is collected, but:

- **Who consents?** If this is a shared home/office, does every occupant have equal say, or only the owner/manager?
- **Ongoing consent:** A notice posted once doesn't capti evolving comfort levels or new occupants who didn't see it initially

傳送訊息

Terms of Use: Deakin GEM is provided to support a range of tasks and activities. By accessing Deakin GEM, you acknowledge and accept that your use is subject to the Student Rights and Responsibilities (for students), the Code of Conduct (for staff), and the Information and Communications Technology Acceptable Use Policy (for all users), along with any related procedures.

Use of responses: Deakin GEM uses artificial intelligence to generate responses. While it aims to provide helpful and relevant information, responses may be incomplete, outdated, or inaccurate and should not be relied on as a sole source of information. Please review and verify information where appropriate.

Privacy Gaps in Your Protocol

1. Consent and awareness asymmetry

Your physical privacy notice informs people *that* data is collected, but:

- **Who consents?** If this is a shared home/office, does every occupant have equal say, or only the owner/manager?
- **Ongoing consent:** A notice posted once doesn't capture *evolving* comfort levels or new occupants who didn't see it initially
- **Power imbalance:** Employees or family members may feel unable to object without social/professional cost

Gap: No mechanism for withdrawing consent or opting out of specific rooms.

2. Scope creep and secondary use

You've defined the current protocol, but:

- Who controls aggregated daily summaries after 90 days? Can they be shared with insurers, landlords, or authorities?
- Could accelerometer data (e.g., patterns of "person standing still at desk for 8 hours") be reused for productivity tracking or behaviour profiling?
- Is there contractual clarity on data deletion after 90 days, or does it persist indefinitely?

Gap: No explicit secondary-use restrictions or data governance policy.

3. Room-level granularity and inference risk

You're not logging *who* is present, but:

- Accelerometer + PIR patterns can infer room occupancy, movement habits, and even *absence patterns* (e.g., "bedroom unoccupied 11 pm–7 am")
- If multiple people share a space, you cannot distinguish them—but aggregated motion patterns still reveal *something* about collective behaviour
- In a family home, this indirectly profiles children, guests, or vulnerable persons

Gap: No de-identification or aggregation safeguard for multi-occupant spaces.

4. Data retention justification

Why 7 days raw + 90 days aggregated?

- Is this driven by theft investigation timelines, or just "seems reasonable"?
- Shorter retention (e.g., 48 hours raw, 30 days summary) would reduce risk

Gap: No documented data minimisation rationale.

Who Else Is Affected?

- **Occupants who didn't consent** (renters, employees, family members)
 - **Guests** (unaware they're being monitored)
 - **Third parties** receiving aggregated data (insurers, property managers, law enforcement if subpoenaed)
 - **Vulnerable groups** (children, elderly, unwell persons whose movement patterns reveal health/safety information)
-

Recommended Additional Safeguard (Privacy-by-Design)

Implement role-based data access with audit logging

Recommended Additional Safeguard (Privacy-by-Design)

Implement role-based data access with audit logging

What:

- Define explicit roles: *device owner, facility manager, security investigator*
- Each role can access only the minimum required:
 - **Owner:** Real-time alerts + 7-day raw logs (if theft suspected)
 - **Facility manager:** Daily aggregates (occupancy trends only)
 - **Investigator:** Full 90-day history (only with documented consent and specific incident)
- Log every access attempt with timestamp, requester, and purpose

Why this aligns with ISO 27001 & Privacy-by-Design:

- **Purpose limitation:** Access tied to defined roles, not open access
- **Accountability:** Audit trail proves who accessed what, when, why
- **Principle of least privilege:** Each user sees only necessary data
- **Transparency:** Occupants can request access logs to see who's viewed *their* data

Implementation:

🔍 搜尋 儲存 複製

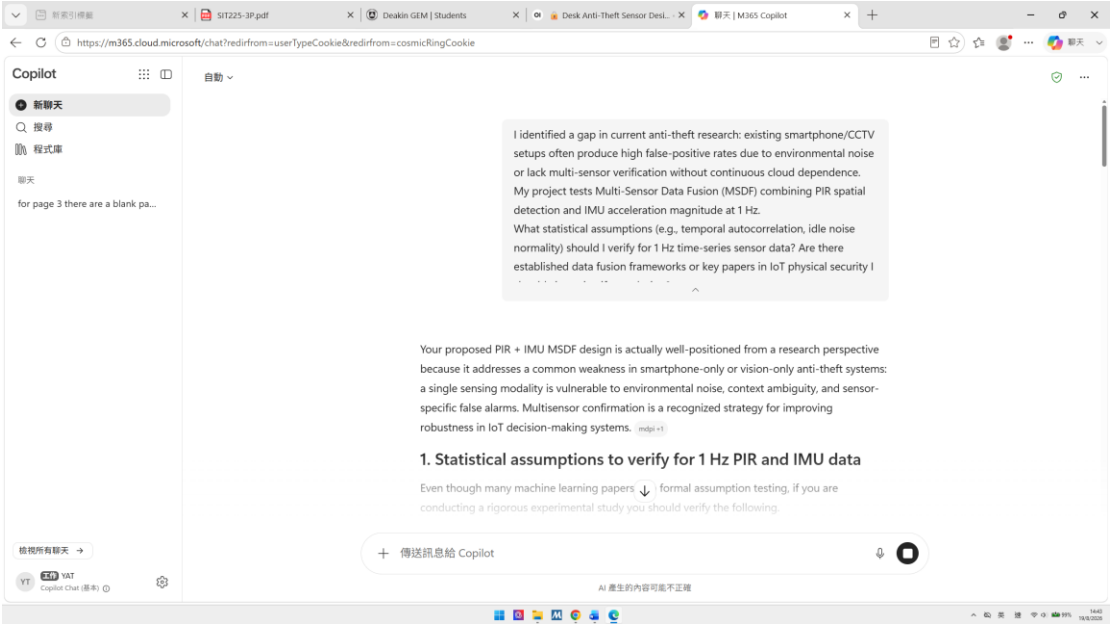
1	Access_Log: [Timestamp, User, Role, Data_Type, Purpose, Approved_By]
2	Example: 2026-08-19 14:30 alice@home Owner 7d_raw theft_incident self

Quick Checklist

- ☐ Document consent mechanism (how do occupants opt out or object?)
- ☐ Define and publish secondary-use policy (what happens to 90-day summaries?)
- ☐ Shorten retention to minimum viable (48 h raw, 30 d aggregate?)
- ☐ Add role-based access control + audit logging
- ☐ Test inference risk: can movement patterns alone reveal health/behaviour?
- ☐ Provide occupants a way to view/request deletion of their data

This shifts you from *notification* to *control*, which is Privacy-by-Design.

Prompt 3 Result: Literature & Methodological Soundness



Your proposed PIR + IMU MSDF design is actually well-positioned from a research perspective because it addresses a common weakness in smartphone-only or vision-only anti-theft systems: a single sensing modality is vulnerable to environmental noise, context ambiguity, and sensor-specific false alarms. Multisensor confirmation is a recognized strategy for improving robustness in IoT decision-making systems. mdpi +1

1. Statistical assumptions to verify for 1 Hz PIR and IMU data

Even though many machine learning papers skip formal assumption testing, if you are conducting a rigorous experimental study you should verify the following.

A. Temporal autocorrelation

At 1 Hz, consecutive observations are rarely independent.

Examples:

- PIR detections often remain active for several consecutive seconds after motion.
- IMU acceleration magnitudes show persistence during movement episodes.

Tests:

- Autocorrelation Function (ACF)
- Partial Autocorrelation Function (PACF)
- Durbin-Watson statistic
- Ljung-Box test

Why important:

If strong autocorrelation exists, treating each second as an independent sample will inflate significance tests and underestimate variance.

B. Stationarity

Security sensors often exhibit changing baselines:

- PIR sensitivity changes with ambient temperature.
- Smartphone IMUs experience orientation drift.
- Environmental conditions vary over time.

Tests:

- Augmented Dickey-Fuller (ADF)

B. Stationarity

Security sensors often exhibit changing baselines:

- PIR sensitivity changes with ambient temperature.
- Smartphone IMUs experience orientation drift.
- Environmental conditions vary over time.

Tests:

- Augmented Dickey-Fuller (ADF)
- KPSS

If non-stationary:

- Difference the series
 - Use rolling baselines
 - Normalize per observation window
-

C. Idle-state noise distribution

For your "device not being stolen" periods, characterize sensor noise.

For IMU magnitude:

$$a_t = \sqrt{x_t^2 + y_t^2 + z_t^2}$$

Check whether idle-state residuals follow:

- Gaussian distribution
- Laplace distribution
- Heavy-tailed distribution

Tests:

- Shapiro-Wilk
- Anderson-Darling
- Q-Q plots

Why useful:

Why useful:

Many threshold-setting methods assume normality.

For example:

$$Threshold = \mu + 3\sigma$$

is only justified when noise is approximately Gaussian.

D. Homoscedasticity (constant variance)

Variance may differ between:

- Day/night
- Indoor/outdoor
- Different device placements

Tests:

- Levene's test
- Bartlett's test

If variance changes substantially, adaptive thresholds are preferable.

E. Sensor synchronization

A key assumption in fusion systems is that PIR and IMU events occur within a meaningful temporal window.

Verify:

- Cross-correlation
- Lag analysis

Example:

If PIR activates 1-2 s before significant IMU acceleration, a fusion rule should account for this lag instead of requiring exact simultaneity.



F. Conditional independence

This is especially important if you use Bayesian fusion.

The naïve assumption is:

$$P(PIR, IMU|Theft) = P(PIR|Theft) \times P(IMU|Theft)$$

In practice, PIR and IMU may become correlated during theft events.

You can evaluate:

- Mutual information
- Pearson/Spearman correlation
- Contingency tables

High dependence suggests more advanced fusion models.

2. Performance metrics you should report

Because your project claims reduced false positives, these metrics are more important than overall accuracy:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2PR}{P + R}$$

Also report:

- False Positive Rate (FPR)
- False Alarm Rate per hour
- Detection latency (seconds)
- ROC-AUC if using probabilistic fusion

For security systems, false alarms are often more important than raw accuracy.

3. Established sensor fusion frameworks you can cite

JDL Data Fusion Model

The Joint Directors of Laboratories (JDL) model is one of the classic sensor-fusion architectures.

It separates fusion into levels:

- Signal/data level
- Feature level
- Decision level

Your PIR + IMU approach naturally fits a feature-level or decision-level fusion architecture. [cite](#)

Bayesian Data Fusion

Good choice when:

- Sensor reliabilities can be estimated.
- Probabilistic outputs are available.

A simple formulation:

$$P(Theft|PIR, IMU) = \frac{P(PIR, IMU|Theft)P(Theft)}{P(PIR, IMU)}$$

This is probably the strongest theoretical justification for a lightweight edge-based anti-theft system.

Dempster-Shafer Evidence Theory

Particularly useful when sensors are uncertain or occasionally conflict.

Advantages:

- Explicitly models uncertainty.
- Handles incomplete information.
- Widely applied in IoT multisensor fusion research. [model + 1](#)

This may fit your work very well because:



- PIR can generate false activations.

- Explicitly models uncertainty.
- Handles incomplete information.
- Widely applied in IoT multisensor fusion research. indpi +1

This may fit your work very well because:

- PIR can generate false activations.
- IMU can experience accidental movement.
- Fusion can combine confidence from both sources.

Kalman Filtering

Kalman filters are the classical approach for continuous-valued sensor fusion and state estimation. They are commonly cited as a standard baseline in multisensor systems. csdn +1

For a 1 Hz anti-theft prototype, however, Bayesian or Dempster-Shafer fusion may be easier to justify than a full state-space model.

4. Relevant papers directly supporting your design

I would strongly recommend citing:

(1) Hamda et al., 2023

"Multisensor Data Fusion in IoT Environments in Dempster-Shafer Theory Setting"

Contributions:

- Discusses uncertainty management in IoT fusion systems.
- Demonstrates improved decision accuracy using evidence-based fusion.
- Provides a strong theoretical rationale for combining heterogeneous sensors. indpi +1

(2) Yuan et al., 2022

"Interpretable Passive Multi-Modal Sensor Fusion for Human Identification and Activity Recognition"

Important because:

- Uses PIR within a multimodal sensing system.
- Shows sensor fusion improves robustness versus single-sensor solutions.
- Emphasizes privacy-preserving alternatives to camera-based monitoring. indpi +1

This is especially relevant if you argue that PIR+IMU reduces dependence on continuous CCTV/cloud processing.

(3) Hall & Llinas

"An Introduction to Multisensor Data Fusion"

A foundational citation commonly used to justify:

- Reduced uncertainty
- Increased reliability
- Better decision-making through complementary sensors

This paper underpins much of the modern JDL fusion literature.

5. A concise research hypothesis

Your methodology could be justified with:

H1: A multi-sensor data fusion framework combining PIR motion detection and smartphone IMU acceleration magnitude at 1 Hz significantly reduces false-positive theft alerts compared with single-sensor detection approaches by requiring corroborating evidence from independent sensing modalities.

That hypothesis aligns well with the established literature on complementary sensor fusion, uncertainty reduction, and decision-level IoT security fusion. indpi +1

For a student research paper, the strongest statistical section would include: **ACF/Ljung-Box** (autocorrelation), **ADF** (stationarity), **Shapiro-Wilk** (idle-noise normality), **cross-correlation** (sensor lag), and **confusion-matrix-based false-positive analysis**. Those tests directly support the validity of a 1 Hz PIR-IMU fusion design.

Part C: Critical Response Report

Introduction

This report discusses the feedback I received from my peer reviewer, Deven Garg, and two AI tools, Deakin GEM and Microsoft Copilot, about my security system. My original design uses an Arduino board, a PIR motion sensor and an LSM6DS3 accelerometer to detect possible desk theft. My aim was to detect more than 95% of theft events while keeping false alarms below 5%. The sensors were originally set to collect data once every second (1 Hz).

The feedback helped me find some problems with my original testing plan. In particular, I needed to improve the sensor reading speed, deal with environmental noise, improve my privacy plan and make sure my data analysis was suitable for the type of data I collected.

Feedback Summary & Thematic Analysis

Theme	Peer feedback	AI feedback	Consensus?
Sampling Rate	pointed out that 1 Hz could miss quick desk bumps that last less than one second.	reading once per second could miss short movements and suggested increasing the rate to around 10–20 times per second.	yes, Both agree 1 Hz is too slow.
Unwanted Noise & Confounders	things such as heat drafts and table vibrations could cause false alarms if they are not controlled.	AI feedback mentioned sunlight, air conditioners and floor vibrations as possible sources of unwanted	Yes, Both agree environmental noise controls are missing.

		sensor readings. It also suggested collecting baseline data first. first.	
Privacy & Ethics	my original design mentioned “facility managers”, which does not really fit a personal desk setup. He also suggested making consent clearer.	movement data could still show information about a person's daily routine, even if no names are recorded.	Yes, Both agree that the privacy section needs changes.
Math & Literature Support	suggested adding more scientific research about security sensors.	checking the statistical methods because sensor readings collected over time may be related to each other. It also suggested using research about multi-sensor systems.	Yes, Both agree that more research and statistical support would be useful.

Four main issues came out of the feedback:

- **Slow Reading Speed:** Both Deven and AI said that 1 Hz is too slow for short events. For example, a quick bump lasting 0.2 seconds could happen between two readings and be missed.
- **Environmental Noise:** Things such as temperature changes, air conditioning and normal vibrations could affect the sensors and cause false alarms.
- **Privacy Problems:** “facility managers” did not fit my home desk situation. The AI feedback also made me think more about how movement timestamps could show when someone is at their desk or away.
- **Statistical Testing:** AI pointed out that readings taken close together in time may not be completely independent. So might need to check this before using some statistical methods.

Identify Conflicts or Tensions

There was some difference between the feedback about how much I should increase the sampling rate.

AI suggested increasing the sensor reading rate to between 10 and 50 times per second. However, I also need to consider the time and hardware limits.

My solution: I think I should increase the accelerometer to 10 Hz on the Arduino. I will still summaries the results every second so that I do not create unnecessarily large amounts of stored data.

Critical Evaluation: Accept / Reject / Revise

Feedback Point	Your decision (Accept / Reject / Revise)	Justification with evidence	Literature / source
Increase reading speed from 1 Hz to 10 Hz	Accept	At 1 Hz, short bumps could be missed. Increasing it to 10 Hz gives the	Deakin GEM AI, Deven Garg

		system more chances to record quick movements.	
Fix heat and draft false alarms	Revise	PIR sensors may react to changes in the environment. But I am not sure how sensitive for PIR to detect environment such as sunlight. Therefore I might need to test under different environment.	
Shorten data storage time	Accept	Keeping raw sensor data for 7 days is more than I need. I will delete the raw data after 48 hours to reduce privacy risks.	ISO 27001, Deakin GEM
Fix wrong user roles	Accept	I removed references to “facility managers” because the system is designed for a personal desk rather than a facility.	Deven Garg
Add statistical independence tests	Accept	Sensor readings collected over time can be	Microsoft Copilot

		related to each other. I will use checks such as ACF and the Ljung-Box test before interpreting the data.	
Add 3 accelerometers across the desk	Reject	Three sensors would add more wiring, testing and setup work than I can reasonably manage. Also don't have enough budget to buy more sensor.	Feasibility limits

1. Faster Sensor Reading Speed (Accept)

Both my peer reviewer and the AI feedback pointed out that 1 Hz is too slow for my experiment. A desk bump can happen very quickly, possibly in less than 0.3 seconds. If the sensor only takes one reading every second, there is a chance that the movement will not be recorded properly.

Because of this, I changed the accelerometer sampling rate from 1 Hz to 10 Hz. This means the board can collect more readings during a short movement and gives me more information to work with.

2. Controlling Environmental Noise (Revise)

The feedback also made me think more about environmental factors. Changes caused by sunlight, heaters or air conditioning could affect the PIR sensor, while the accelerometer may also pick up normal vibrations.

To deal with this, I will collect baseline data before starting the main experiment. This will give me an idea of the normal sensor readings in the room. I can then use this information when setting the alarm thresholds.

3. Better Data Privacy (Accept)

Deven pointed out that my original privacy section included “facility managers”, which does not fit my project because it is designed for a personal. I changed the roles to Main Desk User and Room Occupants.

I also changed how long data is kept. Raw sensor data will now be deleted after 48 hours instead of 7 days. Daily summaries will be kept for 30 days instead of 90 days. This reduces the amount of information stored about people's movements.

4. Math Tests and Sensor Array (Accept & Reject)

Copilot pointed out that sensor readings collected one after another may be related to previous readings. I accepted the suggestion to check this using statistical tests such as autocorrelation.

However, I decided not to use three separate accelerometers. Although this could provide more information about movement across the desk, it would also make the hardware setup and testing more complicated.

Protocol Revision Summary

Original Plan

- **Sensor Reading:** 1 Hz (once per second) for all sensors.
- **Hypotheses:** I originally thought 1 Hz data would be enough to distinguish short bumps from theft using simple averages.
- **Privacy:** Raw data was stored for 7 days and summaries for 90 days. The design also included “facility managers”, which did not fit the intended setup.

Change:

- **Sensor Reading:** The accelerometer will now run at 10 Hz so that quick movements are less likely to be missed. The PIR sensor will remain at 1 Hz.

New Hypotheses:

H1: Combining PIR motion data with 10 Hz accelerometer data will keep false alarms below 5% while maintaining detection above 95%.

H2: Bumps will produce short spikes lasting less than 0.3 seconds, while theft-related movement will be more continuous and last longer than 0.5 seconds. The 10 Hz sampling rate should make this difference easier to identify.

- **Privacy Updates:** The stakeholders are now the desk user and other people sharing the room, such as housemates or coworkers. Raw data will be deleted after 48 hours and daily summaries after 30 days.
- **Analysis Updates:** statistical checks for the time-series data and added research to support the use of multiple sensors.

Reflection: Evaluative Judgement & Professional Growth

The feedback from my peer and the AI tools helped me find problems that I had not noticed in my original design.

At first, I thought 1 Hz was enough because it would reduce memory and power use. After looking at the feedback, I realised that this could actually make it difficult to detect short bumps. I changed the accelerometer to 10 Hz so that more of these quick movements can be recorded.

I also decided not to follow the suggestion of using three accelerometers. Although having more sensors could give me more information, it would make the project harder to build and test within the time limit. This helped me understand that a good research idea also needs to be realistic for the available time and hardware.

The privacy feedback was also useful. I originally focused mainly on names and personal information, but I realised that sensor timestamps can also tell us something about a person's routine. For example, they could show when someone is usually at their desk or away from it. Because of this, I reduced the amount of time that sensor data is stored and changed the user roles in my design. This made me think more carefully about privacy when designing a real-world system.

Acknowledgement of GenAI Use

Per Deakin policy, include an Acknowledgements section:

"This response report was developed with assistance from the following approved Deakin genAI tools:

- Deakin GEM (accessed September 2026): Used to generate structured critique on protocol design and confounding variables (3 iterations).
- Microsoft Copilot for web (accessed September 2026): Used to summarise ASHRAE comfort standards and identify relevant literature.

All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence or peer consensus. The final design, decisions, and justifications are my own."